

ComPort: The ComLab Reservation System for ICS Students of Colegio de Montalban

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ABSTRACT

Introduction: Computer laboratory reservations in academic settings are often handled through manual coordination, which can lead to overlapping schedules, limited visibility of room availability, delayed approvals, and incomplete activity records. **Objective:** This project aimed to design and implement ComPort, a web-based reservation and management system for the computer laboratories of the Institute of Computer Studies at Colegio de Montalban. **Methods:** The project used a descriptive-developmental software engineering approach. The system was built as a full-stack web application utilizing a React frontend, a Node.js/Express backend, and a Prisma ORM with a MySQL database. Key modules include JWT-based authentication, laboratory and schedule management, reservation workflows, PC-level booking, calendar management, reporting, event-driven notifications, and activity logging. **Results:** The completed system provides students with laboratory browsing and reservation capabilities, staff with schedule and reservation oversight, and administrators with broader control over users, laboratories, calendar events, and reports. Conflict prevention is handled through automated schedule validation and booking rules, while dashboard views, automated email reminders, and CSV exports support

operational monitoring. Conclusion: ComPort centralizes laboratory reservation and monitoring tasks into one secure, RESTful platform aligned with the needs of ICS laboratory operations. The system significantly improves visibility, traceability, and strict role-based control compared with a manual reservation process.

INTRODUCTION

Computer laboratories are a core resource for Information and Communication Technology instruction, but access to these rooms is difficult to manage when reservation requests, approvals, and schedule changes are handled manually. Managing physical resources efficiently is a well-documented challenge in educational institutions, where scheduling bottlenecks can hinder academic productivity (**Garg & Kumar, 2020**). In a paper-based setup, users may not immediately see whether a room is available, staff may have to reconcile conflicting bookings by hand, and administrators may have limited visibility over room utilization, maintenance schedules, and reservation history.

For the Institute of Computer Studies at Colegio de Montalban, these limitations create practical problems in day-to-day laboratory coordination. Students need a clear way to request reservations, laboratory staff need a reliable way to review and manage schedules, and administrators need a centralized view of the entire laboratory operation. The shift from manual to digital systems utilizing strict Role-Based Access Control (**RBAC**) is proven to significantly improve security, operational efficiency, and data integrity in organizational environments (**Ferraiolo et al., 2001**). A single system that supports authentication, role-based access, room management, schedule locking, reservation approval, and reporting is therefore useful for improving efficiency and accountability.

ComPort was developed to address these needs. It is a web-based reservation system that allows students to browse laboratories, choose an available schedule, and request either a whole-laboratory reservation or a specific PC reservation. Laboratory staff and administrators can manage laboratories, maintain schedules, review reservations, monitor activity, and generate reports. The project is intended to replace fragmented or manual coordination with a more organized, architecturally sound, and auditable workflow.

METHODS

Context and Setting of the Study This project was developed for the computer laboratory environment of the Institute of Computer Studies at Colegio de Montalban. The operational context includes students who request laboratory access, laboratory staff who manage room availability and reservation approvals, and administrators who oversee users, laboratories, schedules, calendar blocks, and reports.

Study Design The project followed a descriptive-developmental software design. Rather than testing a medical or behavioral intervention, the work focused on planning, building, and validating a functional information system that automates laboratory reservation and management workflows.

Population The intended user population consists of:

1. ICS students who request reservations.
2. Laboratory staff who manage schedules, review requests, and monitor room usage.
3. Administrators who manage system users, laboratory records, calendar events, and reports.

Sampling Strategy No probabilistic sampling strategy was used because this is a system development project rather than a population survey. User roles and workflow requirements were derived from the operational needs of the laboratory environment and encoded into the system design.

Intervention The intervention was the implementation of ComPort as a centralized reservation and management platform. The technological architecture utilizes a React.js frontend built with Vite and Tailwind CSS, coupled with a Node.js and Express.js backend. Data persistence is handled by a MySQL database managed through the Prisma Object-Relational Mapper (**ORM**). The implemented system features include:

1. Student registration and secure login utilizing JSON Web Tokens (**JWT**) and bcrypt password hashing.
2. Role-based routing and dashboards for students, laboratory staff, and administrators.

3. Laboratory management with capacity definitions, custodian assignments, and laboratory images.
4. Schedule management with automated overlap prevention logic.
5. Reservation creation for either whole-laboratory or granular PC-specific bookings.
6. Reservation review workflows (approval, rejection, cancellation, and completion).
7. Management calendar features for maintenance and holiday blocking.
8. Event-driven asynchronous notifications (NotificationEventBus) and automated email reminders (ReservationReminderService).
9. Reporting mechanisms, CSV export capabilities, and continuous activity logging (ActivityLogService).

Main Study Variables The main system outcomes of interest were:

1. Reservation efficiency and schedule conflict prevention algorithms.
2. Implementation of Role-Based Access Control (RBAC) and data visibility.
3. Traceability of reservation decisions and system auditability.

Data Collection Instruments and Procedures This paper is based on the completed codebase, database schema, RESTful API structure, and user interface modules of the project. During development, functional modules were strictly organized by role middleware (e.g., requireRole) so that students, staff, and administrators only access the endpoints relevant to their permissions.

Outline of Analysis Methods The analysis was descriptive and implementation-based. System behavior was examined through feature-to-requirement matching, review of role-based workflow coverage via unit and integration tests (using Vitest), inspection of Prisma schema validation logic, and verification of reporting, logging, and asynchronous task execution capabilities.

RESULTS

Report on Data Collection and/or Recruitment The system was successfully developed, containerized using Docker, and populated with implementation data representing the ICS

computer laboratories, schedules, and test accounts. Because this phase focused on software engineering and structural validation, formal recruitment of human subjects for a live operational trial was outside the scope of this initial rollout.

Participants (Demographic, Clinical Condition, etc.) The system accounts are structured around three distinct participant profiles defined in the database schema:

1. Students: Users who register, browse availability via a weekly schedule grid, and submit reservation requests.
2. Laboratory Staff: Users assigned to specific laboratories, responsible for managing records, reviewing requests, and monitoring daily statuses.
3. Administrators: Users with global oversight who manage staff assignments, all users, and the centralized management calendar.

Present Key Findings with Respect to the Central Research Question The project successfully produced a working web-based reservation system that programmatically enforces conflict prevention. The backend booking workflow correctly checks selected schedules, queries the Prisma database to validate time ranges, and rejects overlapping reservations before they are committed. The reservation interface successfully supports two distinct operational modes: reserving the entire laboratory or reserving a specific PC within a laboratory.

Role-based interfaces and API routes were successfully isolated. The staff interface effectively provides summary metric cards, reservation status filters, dynamic charts, and CSV data exports to support daily oversight. Administrators are provided a unified calendar view that seamlessly merges custom calendar entries (maintenance/holiday blocks) with derived items from schedules and reservations, centralizing operational visibility into a single dashboard.

Secondary Findings The backend architectural schema robustly handles secondary operational requirements. Entities such as password reset tokens, in-app notifications, and comprehensive audit trails (Activity Logs) operate successfully to ensure system accountability and secure user recovery. Furthermore, the system incorporates automated integration testing, global error handling, API rate limiting, and production-oriented environment configuration,

which collectively strengthen the application against common web vulnerabilities and prepare it for deployment.

Discussion

Main Findings of the Study ComPort addresses the major weaknesses of manual laboratory reservation by successfully consolidating scheduling, approval, monitoring, and automated notifications into one digital platform. The most important contribution of the system is the way it structures the reservation workflow around strict, programmatic role-based permissions and database-level conflict prevention. This heavily reduces the human error associated with duplicate bookings and makes it significantly easier for staff to view the real-time status of their assigned resources.

Discuss the Main Results with Reference to Previous Research Consistent with the broader direction of existing academic laboratory management systems, ComPort emphasizes centralized access and better accountability (**Garg & Kumar, 2020**). The adoption of a strict Role-Based Access Control (**RBAC**) model follows established information security standards, ensuring that data modification is restricted strictly to authorized personnel (**Ferraiolo et al., 2001**). Compared with manual logbooks or informal coordination workflows previously utilized, ComPort provides a highly visible and verifiable audit trail through digital reservation records, automated activity logs, and a centralized notification event bus.

Policy and Practice Implications of the Results For the ICS department at Colegio de Montalban, this system implies a transition toward data-driven laboratory management. The reporting tools and CSV exports add direct practical value by helping staff identify peak reservation patterns and laboratory demand. This allows department heads to make informed policy decisions regarding laboratory operating hours, PC hardware maintenance schedules, and budget resource allocation based on actual usage metrics rather than estimation.

Strengths and Limitations of the Study A major strength of the system is its modern, decoupled architectural style (**Fielding, 2000**), separating concerns cleanly across the Express backend API, the React frontend, and the database schema. It also introduces granular PC-level reservation support, which is much more flexible than traditional room-only booking models. The

main limitation is that this study does not currently report a formal field experiment or statistically analyzed user survey. Consequently, the paper confidently describes the implemented technical capabilities but cannot claim quantitatively measured improvements in user satisfaction or manual processing time.

CONCLUSIONS

ComPort is a highly functional, full-stack web-based reservation system that centralizes computer laboratory scheduling, approval, monitoring, and reporting for the Institute of Computer Studies at Colegio de Montalban. The system vastly improves visibility over laboratory availability, enforces robust role-based access control, and heavily mitigates reservation conflicts through automated backend validation. Its underlying architecture is scalable and highly appropriate for academic laboratory operations.

Future work should include a formal usability study and statistical evaluation to measure user satisfaction and time-savings compared to the manual process. Additional development perspectives for future iterations include QR-based attendance or sign-in support for verified usage tracking, deeper predictive analytics for resource allocation, and mobile-responsive enhancements for faster, on-the-go access via campus devices.

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REFERENCES

- Adewasi, D. (2021, June 14). What is IMRaD? IMRaD format in simple terms! Scientific-editing.info.
- Cuschieri, S., Grech, V., & Savona-Ventura, C. (2019). WASP (Write a scientific paper): Structuring a scientific paper. *Early Human Development*, 128, 114-117. <https://doi.org/10.1016/j.earlhumdev.2018.09.011>
- Ferraiolo, D. F., Sandhu, R., Gavrila, S., Kuhn, D. R., & Chandramouli, R. (2001). Proposed NIST standard for role-based access control. *ACM Transactions on Information and System Security (TISSEC)*, 4(3), 224-274. <https://doi.org/10.1145/501978.501980>
- Fielding, R. T. (2000). Architectural styles and the design of network-based software architectures [Doctoral dissertation, University of California, Irvine].
- Garg, R., & Kumar, S. (2020). Automated scheduling and resource management in academic institutions. *Journal of Educational Technology Systems*, 48(4), 543-562.
- Nair, P. K. R., & Nair, V. D. (2014). Organization of a research paper: The IMRAD format. In *Scientific Writing and Communication in Agriculture and Natural Resources*. Springer. https://doi.org/10.1007/978-3-319-03101-9_2
- Sollaci, L. B., & Pereira, M. G. (2004). The introduction, methods, results, and discussion (IMRAD) structure: A fifty-year survey. *Journal of the Medical Library Association*, 92(3), 364-367.